

Therapeutic applications of aptamers.

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Aptamers constitute a new class of oligonucleotides that have gained therapeutic importance. With the approval of the first aptamer drug, pegaptanib, interest in this class of oligonucleotides, often referred to as 'chemical antibodies', has increased. This article discusses aptamers in relation to other oligonucleotide molecules such as antisense nucleotides, short inhibitory sequences, ribozymes and so on. The development of pegaptanib is looked at from the point of view of the challenges faced in converting aptamers into therapeutic molecules. Cases of other aptamers, which show promise as drugs, are discussed in slightly greater detail. Comparison with antibodies and small molecules, which have hitherto held monopoly in this area, is also made.

Selection of smart aptamers by methods of kinetic capillary electrophoresis.

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We coin the term "smart aptamers" -- aptamers with predefined binding parameters (k_{on}), k_{off}), K_d) of aptamer-target interaction. Aptamers, in general, are oligonucleotides, which are capable of binding target molecules with high affinity and selectivity. They are considered as potential therapeutic targets and also thought to rival antibodies in immunoassay-like analyses. Aptamers are selected from combinatorial libraries of oligonucleotides by affinity methods. Until now, technological limitations have precluded the development of smart aptamers. Here, we report on two kinetic capillary electrophoresis techniques applicable to the selection of smart aptamers. Equilibrium capillary electrophoresis of equilibrium mixtures was used to develop aptamers with predefined equilibrium dissociation constants (K_d), while nonequilibrium capillary electrophoresis of equilibrium mixtures facilitated selection of aptamers with different dissociation rate constants (k_{off}). Selections were made for MutS protein, for which aptamers have never been previously developed. Both theoretical and practical aspects of smart aptamer development are presented, and the advantages of this new type of affinity probes are described.

Aptamers in the virologists' toolkit.

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Aptamers are artificial nucleic acid ligands that can be generated in vitro against a wide range of molecules, including the gene products of viruses. Aptamers are isolated from complex libraries of synthetic nucleic acids by an iterative, cell-free process that involves repetitively reducing the complexity of the library by partitioning on the basis of selective binding to the target molecule, followed by reamplification. For virologists, aptamers have potential uses as tools to help to analyse the molecular biology of virus replication, as a complement to the more familiar monoclonal antibodies. They also have potential applications as diagnostic biosensors and in the development of antiviral agents. In recent years, these two promising avenues have been explored increasingly by virologists; here, the progress that has been made is reviewed.

Oligonucleotide aptamers: new tools for targeted cancer therapy.

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Aptamers are a class of small nucleic acid ligands that are composed of RNA or single-stranded DNA oligonucleotides and have high specificity and affinity for their targets. Similar to antibodies, aptamers interact with their targets by recognizing a specific three-dimensional structure and are thus termed "chemical antibodies." In contrast to protein antibodies, aptamers offer unique chemical and biological characteristics based on their oligonucleotide properties. Hence, they are more suitable for the development of novel clinical applications. Aptamer technology has been widely investigated in various biomedical fields for biomarker discovery, in vitro diagnosis, in vivo imaging, and targeted therapy. This review will discuss the potential applications of aptamer technology as a new tool for targeted cancer therapy with emphasis on the development of aptamers that are able to specifically target cell surface biomarkers. Additionally, we will describe several approaches for the use of aptamers in targeted therapeutics, including aptamer-drug conjugation, aptamer-nanoparticle conjugation, aptamer-mediated targeted gene therapy, aptamer-mediated immunotherapy, and aptamer-mediated biotherapy.

The development and testing of aptamers for cancer.

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Aptamers are single-stranded DNA or RNA oligonucleotides that have specific three-dimensional structures and bind to target molecules. Several unique properties of aptamers, including high binding specificity, low immunogenicity, structural stability and ease of synthesis, have made aptamers promising agents for directed therapy against cancer targets. Aptamers with anti-neoplastic activity against extracellular, cell membrane and intracellular targets have been developed. Aptamers also have been incorporated into novel constructs as tools for delivering therapeutic agents. This review discusses the most recent advances in the application of aptamer technology for cancer therapy.

Designed and Evolved Nucleic Acid Nanotechnology: Contrast and Complementarity.

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In this review, we explore progress on DNA aptamers (evolved DNA), DNA circuits (designed DNA), and the newest projects that integrate both. Designed DNA nanotechnology includes static nanostructures, dynamic nanodevices, and reaction networks (sometimes called DNA circuits). DNA circuits are dynamic DNA reactions that perform computations and sequence-specific amplification. Directed evolution can be used to produce DNA that can recognize specific targets. Aptamers are evolved nucleic acids; they are produced artificially with an in vitro selection process. DNA aptamers are molecular recognition elements made of single-stranded DNA (ssDNA) with the potential to interact with proteins, small molecules, viruses, and even cells. Designed molecular structures can incorporate aptamers for applications with immediate practical impact.

Aptamers and their applications in nanomedicine.

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Aptamers are composed of short RNA or single-stranded DNA sequences that, when folded into their unique 3D conformation, can bind to their targets with high specificity and affinity. Although functionally similar to protein antibodies, oligonucleotide aptamers offer several advantages over protein antibodies in biomedical and clinical applications. Through the enhanced permeability and retention effect, nanomedicines can improve the therapeutic index of a treatment and reduce side effects by enhancing accumulation at the disease site. However, this targets tumors passively and, thus, may not be ideal for targeted therapy. To construct ligand-directed "active targeting" nanobased delivery systems, aptamer-equipped nanomedicines have been tested for in vitro diagnosis, in vivo imaging, targeted cancer therapy, theranostic approaches, sub-cellular molecule detection, food safety, and environmental monitoring. This review focuses on the development of aptamer-conjugated nanomedicines and their application for in vivo imaging, targeted therapy, and theranostics.

Therapeutic aptamers: developmental potential as anticancer drugs.

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Aptamers, composed of single-stranded DNA or RNA oligonucleotides that interact with target molecules through a specific three-dimensional structure, are selected from pools of combinatorial oligonucleotide libraries. With their high specificity and affinity for target proteins, ease of synthesis and modification, and low immunogenicity and toxicity, aptamers are considered to be attractive molecules for development as anticancer therapeutics. Two aptamers - one targeting nucleolin and a second targeting CXCL12 - are currently undergoing clinical trials for treating cancer patients, and many more are under study. In this mini-review, we present the current clinical status of aptamers and aptamer-based cancer therapeutics. We also discuss advantages, limitations, and prospects for aptamers as cancer therapeutics.

Generation of Specific Aptamers.

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Nucleic acid aptamers are therapeutic agents consisting of short single-strand DNA or RNA oligonucleotides, which have the ability to bind to target therapeutic molecules with high affinity and specificity and have been developed as potent drugs for the treatment of rheumatoid arthritis. Aptamers have unique and advantageous features over antibodies, such as superior affinity with nano- or pico-molar dissociation constants and ease of chemical synthesis, modification, and inactivation by designing antisense sequences. In this chapter, using a DNA-oligonucleotide pool, the technology of proteoliposome-systematic evolution of ligands by exponential enrichment (SELEX) is introduced. By using this technique, potential therapeutic agents with high affinity and specificity could be obtained.

Aptamers as Therapeutics.

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Aptamers are single-stranded nucleic acid molecules that bind to and inhibit proteins and are commonly produced by systematic evolution of ligands by exponential enrichment (SELEX). Aptamers undergo extensive pharmacological revision, which alters affinity, specificity, and therapeutic half-life, tailoring each drug for a specific clinical need. The first therapeutic aptamer was described 25 years ago. Thus far, one aptamer has been approved for clinical use, and numerous others are in preclinical or clinical development. This review presents a short history of aptamers and SELEX, describes their pharmacological development and optimization, and reviews potential treatment of diseases including visual disorders, thrombosis, and cancer.
